SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool : Pseudocode Writing Task (Algorithm Design)
Weightage: 20%

QUESTIONS: **Total Marks: 25**

1: XML Parsing Algorithm

Write a detailed **pseudocode algorithm** to parse an XML document that contains structured data such as user or product details. Your algorithm should clearly describe the steps involved in loading the XML file, accessing its elements, traversing through nodes, extracting required data, and displaying the output in a readable format. Ensure that the logic handles hierarchical structure properly.

2: MVC Workflow

Write a **pseudocode representation** of the MVC (Model–View–Controller) workflow in a web application. Your answer should clearly illustrate how a user request is received, processed by the Controller, how the Model interacts with the database, and how the View generates and returns the response to the user. Include all major steps involved in request handling and response generation.

3: JSP Request Flow

Write a detailed **pseudocode algorithm** to explain the request–response flow in a JSP-based web application. Your algorithm should include steps such as client request initiation, server processing, interaction with backend components (like Servlets or database), and generation of dynamic content that is sent back to the client.

4: Database Retrieval

Write a **pseudocode algorithm** to retrieve data from a database. Your answer should include steps for establishing a database connection, executing SQL queries, processing the result set, and displaying the retrieved data. Clearly represent the sequence of operations involved in database interaction.

5: PHP Validation

Write a **pseudocode algorithm** for validating user input in a web form using PHP concepts. The algorithm should include checks for empty fields, validation of input formats (such as email and password), handling invalid inputs with appropriate error messages, and allowing submission only when all validations are satisfied.

Code of Conduct:

I VARSHA S 192462003 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1. XML Parsing

```
START
Load XML file
Check XML validity
Get root node
FOR each child node
    Read elements
    Extract values
    Display data
END FOR
STOP
```

2. MVC Workflow

```
START
User sends request
Controller receives request
Controller calls Model
Model accesses database
Model returns data
Controller sends data to View
View generates response
Display response to User
STOP
```

3. JSP Request Flow

```
START
Client sends request
Server receives request
JSP is processed
Backend/Servlet is called
Database data is retrieved
JSP creates HTML
Server sends response
Client displays page
STOP
```

4. Database Retrieval

```
START
Connect to database
Create SELECT query
Execute query
Read ResultSet
Display records
Close ResultSet
Close connection
STOP
```

5. PHP Validation

```
START
Receive form data
Check empty fields
Validate email
Validate password
IF valid
    Accept and process data
ELSE
    Display error message
END IF
STOP
```

Result

The required pseudocode algorithms were successfully designed for XML parsing, MVC workflow, JSP request flow, database retrieval and PHP validation.